Real Data Analysis

```
## [1] "The sample size is 7585."
```

```
## [1] "The exposure prevalence is 0.448."
```

```
## [1] "The outcome prevalence is 0.237."
```

Method 1: Kitchen Sink

```
## Waiting for profiling to be done...
## Waiting for profiling to be done...
```

```
## [1] "The number of proxy variables selected by kitchen sink method is 142."
```

```
## [1] "The Odds Ratio by Ks formula method: "
```

```
##    OR    SE lower upper
## 1.502 0.038 0.333 0.481
```

```
## [1] "The Risk Difference by Ks formula method: "
```

```
##    RD    SE lower upper
## 0.075 0.015 0.056 0.095
```

Method 2: Bross Formula

```
## Waiting for profiling to be done...
## Waiting for profiling to be done...
```

```
## [1] "The number of proxy variables selected by bross formula method is 100."
```

```
## [1] "The Odds Ratio by Bross formula method: "
```

```
##    OR    SE lower upper
## 1.522 0.038 0.346 0.495
```

```
## [1] "The Risk Difference by Bross formula method: "
```

```
##    RD    SE lower upper
## 0.077 0.014 0.058 0.096
```

Method 3: Hybrid Bross Formula & Lasso

```
## Waiting for profiling to be done...
## Waiting for profiling to be done...

## [1] "The number of proxy variables selected by hybrid method is 49."

## [1] "The Odds Ratio by Hybrid formula method: "

##    OR    SE lower upper
## 1.543 0.038 0.359 0.509

## [1] "The Risk Difference by Hybrid formula method: "

##    RD    SE lower upper
## 0.079 0.013 0.060 0.099
```

```
hdps.dim <- out2

load("../data/analytic3cycles.RData")
hdps.dim$id <- hdps.dim$idx
hdps.dim$idx <- NULL
hdps.data <- merge(data.complete[,c("id",
                                    outcome,
                                    exposure,
                                    investigator.specified.covariates)],
                   hdps.dim, by = "id")

hdps.data$exposure <- as.numeric(I(hdps.data$obese=='Yes'))
hdps.data$outcome <- as.numeric(I(hdps.data$diabetes=='Yes'))

proxy.list <- names(out2[,-1])
proxyform <- paste0(proxy.list, collapse = "+")
covform <- paste0(investigator.specified.covariates, collapse = "+")
rhsformula <- paste0(c(covform, proxyform), collapse = "+")
ps.formula <- as.formula(paste0("exposure", "~", rhsformula))
covarsTfull <- c(investigator.specified.covariates, proxy.list)
Y.form <- as.formula(paste0(c("outcome~ exposure",
                              covarsTfull), collapse = "+") )
full.formula <- as.formula(paste0("outcome~exposure+",
                                  rhsformula,
                                  collapse = "+"))
```

Method 4: Pure Lasso

```
set.seed(42)
start_time <- Sys.time()
covar.mat <- model.matrix(Y.form, data = hdps.data)[,-1]
lasso.fit <- glmnet::cv.glmnet(y = hdps.data$outcome,
                               x = covar.mat,
                               type.measure='mse',
                               family="binomial",
                               alpha = 1,
                               nfolds = 5)
```

```r
coef.fit <- coef(lasso.fit,s='lambda.min',exact=TRUE)
sel.variables <- row.names(coef.fit)[which(as.numeric(coef.fit)!=0)]
proxy_lasso <- proxy.list[proxy.list %in% sel.variables]
proxyform <- paste0(proxy_lasso, collapse = "+")
rhsformula <- paste0(c(covform, proxyform), collapse = "+")
ps.formula <- as.formula(paste0("exposure", "~", rhsformula))

W.out_lasso <- weightit(ps.formula,
                  data = hdps.data,
                  estimand = "ATE",
                  method = "ps")
fit.OR_lasso <- glm(out.formula,
           data = hdps.data,
           weights = W.out_lasso$weights,
           family= binomial(link = "logit"))
sum.OR_lasso <- summary(fit.OR_lasso)$coef["exposure",
                                 c("Estimate",
                                   "Std. Error",
                                   "Pr(>|z|)")]
ci.OR_lasso <- confint(fit.OR_lasso, level = 0.95)["exposure", ]
```

```
## Waiting for profiling to be done...
```

```r
sum.ci.OR_lasso <- c(sum.OR_lasso, ci.OR_lasso)
sum.ci.OR_lasso <- c(exp(sum.ci.OR_lasso[1]), sum.ci.OR_lasso[2], sum.ci.OR_lasso[4:5])
names(sum.ci.OR_lasso) <- c("Odds Ratio", "Std. Error", "2.5 %", "97.5 %")
names(sum.ci.OR_lasso) <- c("OR", "SE", "lower", "upper")

fit.RD_lasso <- glm(out.formula,
                     data= hdps.data,
                     weights= W.out_lasso$weights,
                     family=gaussian(link= "identity"))
sum.RD_lasso  <- summary(fit.RD_lasso)$coef["exposure",
                                        c("Estimate",
                                          "Std. Error",
                                          "Pr(>|t|)")]
sum.RD_lasso[2]<- sqrt(sandwich::sandwich(fit.RD_lasso)[2,2])
ci.RD_lasso <- confint(fit.RD_lasso, "exposure", level = 0.95,  method = "hc1")
```

```
## Waiting for profiling to be done...
```

```r
sum.ci.RD_lasso  <- c(sum.RD_lasso[1:2], ci.RD_lasso)
names(sum.ci.RD_lasso) <- c("RD", "SE", "lower", "upper")

paste0("The number of proxy variables selected by lasso method is ", length(proxy_lasso), ".")
```

```
## [1] "The number of proxy variables selected by lasso method is 60."
```

```r
print("The Odds Ratio by Lasso formula method: ")
```

```
## [1] "The Odds Ratio by Lasso formula method: "
```

```r
round(sum.ci.OR_lasso, 3)
```

```
##    OR    SE lower upper
## 1.531 0.038 0.352 0.501
```

```r
print("The Risk Difference by Lasso formula method: ")
```

```
## [1] "The Risk Difference by Lasso formula method: "
```

```r
round(sum.ci.RD_lasso, 3)
```

```
##    RD    SE lower upper
## 0.078 0.014 0.059 0.098
```

```r
end_time <- Sys.time()
elapsed.time_lasso <- as.numeric(difftime(end_time, start_time, units = "secs"))
```

Method 5: Elastic Net

```
## Waiting for profiling to be done...
## Waiting for profiling to be done...
```

```
## [1] "The number of proxy variables selected by elastic net model is 69."
```

```
## [1] "The Odds Ratio by Enet formula method: "
```

```
##    OR    SE lower upper
## 1.504 0.038 0.334 0.482
```

```
## [1] "The Risk Difference by Enet formula method: "
```

```
##    RD    SE lower upper
## 0.075 0.014 0.056 0.094
```

Method 6: Random Forest

```
## Waiting for profiling to be done...
## Waiting for profiling to be done...
```

```
## [1] "The number of proxy variables selected by random forest is 100."
```

```
## [1] "The Odds Ratio by Rf formula method: "
```

```
##    OR    SE lower upper
## 1.551 0.038 0.364 0.514
```

```
## [1] "The Risk Difference by Rf formula method: "
```

```
##    RD    SE lower upper
## 0.080 0.013 0.061 0.099
```

Method 7: XGBoost

```
## [1]  train-logloss:0.295926
## [2]  train-logloss:0.184700
## [3]  train-logloss:0.127870
## [4]  train-logloss:0.092927
## [5]  train-logloss:0.070237
```

```
## Waiting for profiling to be done...
## Waiting for profiling to be done...
```

```
## [1] "The number of proxy variables selected by elastic net model is 48."
```

```
## [1] "The Odds Ratio by Xgboost formula method: "
```

```
##    OR    SE lower upper
## 1.564 0.038 0.373 0.522
```

```
## [1] "The Risk Difference by Xgboost formula method: "
```

```
##    RD    SE lower upper
## 0.082 0.013 0.063 0.101
```

Method 8.1: Stepwise - Forward Selection

```
## Waiting for profiling to be done...
## Waiting for profiling to be done...
```

```
## [1] "The number of proxy variables selected by elastic net model is 59."
```

```
## [1] "The Odds Ratio by Forward formula method: "
```

```
##    OR    SE lower upper
## 1.533 0.038 0.353 0.502
```

```
## [1] "The Risk Difference by Forward formula method: "
```

```
##    RD    SE lower upper
## 0.078 0.013 0.059 0.097
```

Method 8.2: Stepwise - Backward Elimination

```
## Waiting for profiling to be done...
## Waiting for profiling to be done...
```

```
## [1] "The number of proxy variables selected by elastic net model is 59."
```

```
## [1] "The Odds Ratio by Backward formula method: "

##    OR    SE lower upper
## 1.533 0.038 0.353 0.502

## [1] "The Risk Difference by Backward formula method: "

##    RD    SE lower upper
## 0.078 0.013 0.059 0.097
```

Method 9: Genetic algorithm (GA)

```
## Waiting for profiling to be done...
## Waiting for profiling to be done...

## [1] "The number of proxy variables selected by elastic net model is 64."

## [1] "The Odds Ratio by Ga formula method: "

##    OR    SE lower upper
## 1.515 0.038 0.341 0.490

## [1] "The Risk Difference by Ga formula method: "

##    RD    SE lower upper
## 0.076 0.014 0.057 0.096

##              elapsed_time
## kitchen sink     4.120014
## bross            2.857916
## hybird           3.281890
## lasso            4.082564
## enet             4.617078
## rf             565.524614
## xgboost          1.448758
## forward          5.460790
## backward         9.510116
## ga             127.483694
```

Comparing the Results:

```
methods <- c("kitchen sink", "bross", "hybrid",
             "lasso", "enet",
             "rf", "xgboost",
             "forward", "backward",
             "ga")

num.proxy_all <- data.frame(num.of.proxy = c(length(proxy_ks),
                                             length(proxy_bross),
                                             length(proxy_hybrid),
```

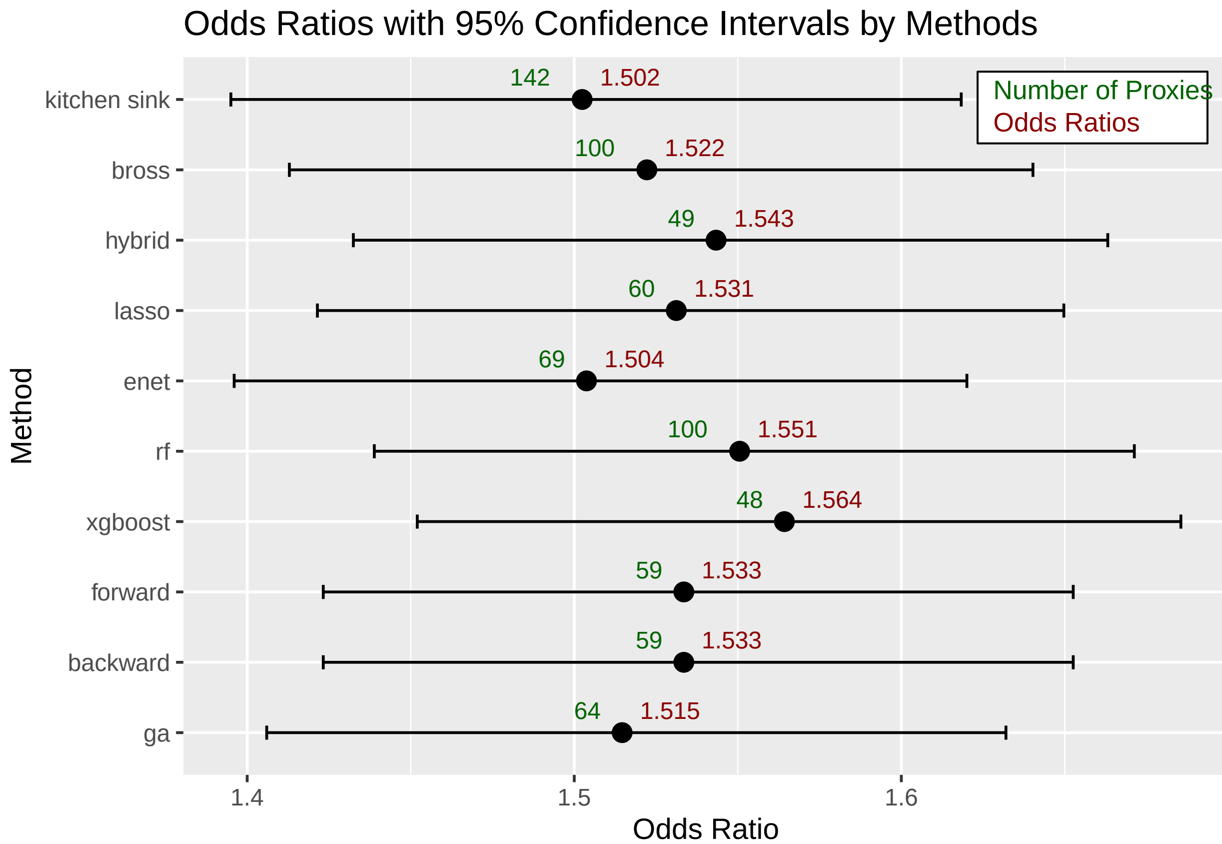
```r
                                        length(proxy_lasso),
                                        length(proxy_enet),
                                        length(proxy_rf),
                                        length(proxy_xgboost),
                                        length(proxy_forward),
                                        length(proxy_backward),
                                        length(proxy_ga)),
                           num.proxy.in.common = c(NA,
                                                   length(proxy_bross[proxy_bross %in% proxy_bross]),
                                                   length(proxy_hybrid[proxy_hybrid %in% proxy_bross])
                                                   length(proxy_lasso[proxy_lasso %in% proxy_bross]),
                                                   length(proxy_enet[proxy_enet %in% proxy_bross]),
                                                   length(proxy_rf[proxy_rf %in% proxy_bross]),
                                                   length(proxy_xgboost[proxy_xgboost %in% proxy_bross
                                                   length(proxy_forward[proxy_forward %in% proxy_bross
                                                   length(proxy_backward[proxy_backward %in% proxy_bro
                                                   length(proxy_ga[proxy_ga %in% proxy_bross])))
rownames(num.proxy_all) <-  methods

num.proxy_all$pct.in.common <- num.proxy_all$num.proxy.in.common / num.proxy_all$num.of.proxy

proxy.common.1to1 <- data.frame(ini = rep(NA, 10))

for (i in 1:10) {
  col_name <- methods[i]
  na.i <- 1
  na <- c()
  while (na.i < i) {
    na <- c(na, NA)
    na.i <- na.i + 1
  }
  if (col_name == "kitchen sink") {
    col_name <- "ks"
  }
  num_common <- c(length(proxy_ks[proxy_ks %in% get(paste0("proxy_",col_name))]),
                  length(proxy_bross[proxy_bross %in% get(paste0("proxy_",col_name))]),
                  length(proxy_hybrid[proxy_hybrid %in% get(paste0("proxy_",col_name))]),
                  length(proxy_lasso[proxy_lasso %in% get(paste0("proxy_",col_name))]),
                  length(proxy_enet[proxy_enet %in% get(paste0("proxy_",col_name))]),
                  length(proxy_rf[proxy_rf %in% get(paste0("proxy_",col_name))]),
                  length(proxy_xgboost[proxy_xgboost %in% get(paste0("proxy_",col_name))]),
                  length(proxy_forward[proxy_forward %in% get(paste0("proxy_",col_name))]),
                  length(proxy_backward[proxy_backward %in% get(paste0("proxy_",col_name))]),
                  length(proxy_ga[proxy_ga %in% get(paste0("proxy_",col_name))]))
  proxy.common.1to1[i] <- c(na, num_common[i:10])
}

colnames(proxy.common.1to1) <- rownames(num.proxy_all)
rownames(proxy.common.1to1) <- rownames(num.proxy_all)

num.proxy_all
```

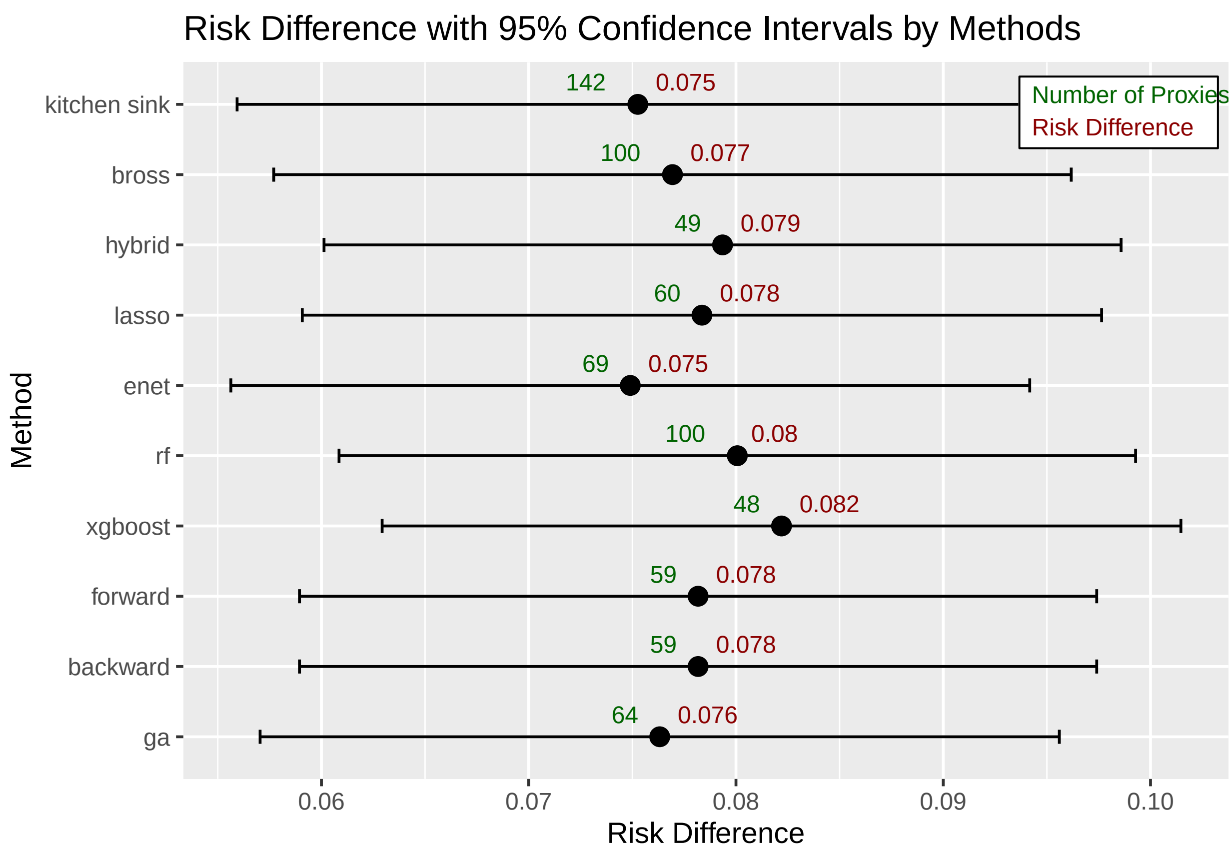
```
##                num.of.proxy num.proxy.in.common pct.in.common
```

```
## kitchen sink          142                    NA          NA
## bross                 100                   100   1.0000000
## hybrid                 49                    49   1.0000000
## lasso                  60                    47   0.7833333
## enet                   69                    54   0.7826087
## rf                    100                    68   0.6800000
## xgboost                48                    38   0.7916667
## forward                59                    45   0.7627119
## backward               59                    45   0.7627119
## ga                     64                    44   0.6875000
```

```
proxy.common.1to1
```

```
##              kitchen sink bross hybrid lasso enet  rf xgboost forward backward
## kitchen sink          142    NA     NA    NA   NA  NA      NA      NA       NA
## bross                 100   100     NA    NA   NA  NA      NA      NA       NA
## hybrid                 49    49     49    NA   NA  NA      NA      NA       NA
## lasso                  60    47     47    60   NA  NA      NA      NA       NA
## enet                   69    54     48    60   69  NA      NA      NA       NA
## rf                    100    68     32    41   48 100      NA      NA       NA
## xgboost                48    38     24    28   30  29      48      NA       NA
## forward                59    45     41    51   54  38      25      59       NA
## backward               59    45     41    51   54  38      25      59       59
## ga                     64    44     28    36   40  49      25      35       35
##              ga
## kitchen sink NA
## bross        NA
## hybrid       NA
## lasso        NA
## enet         NA
## rf           NA
## xgboost      NA
## forward      NA
## backward     NA
## ga           64
```

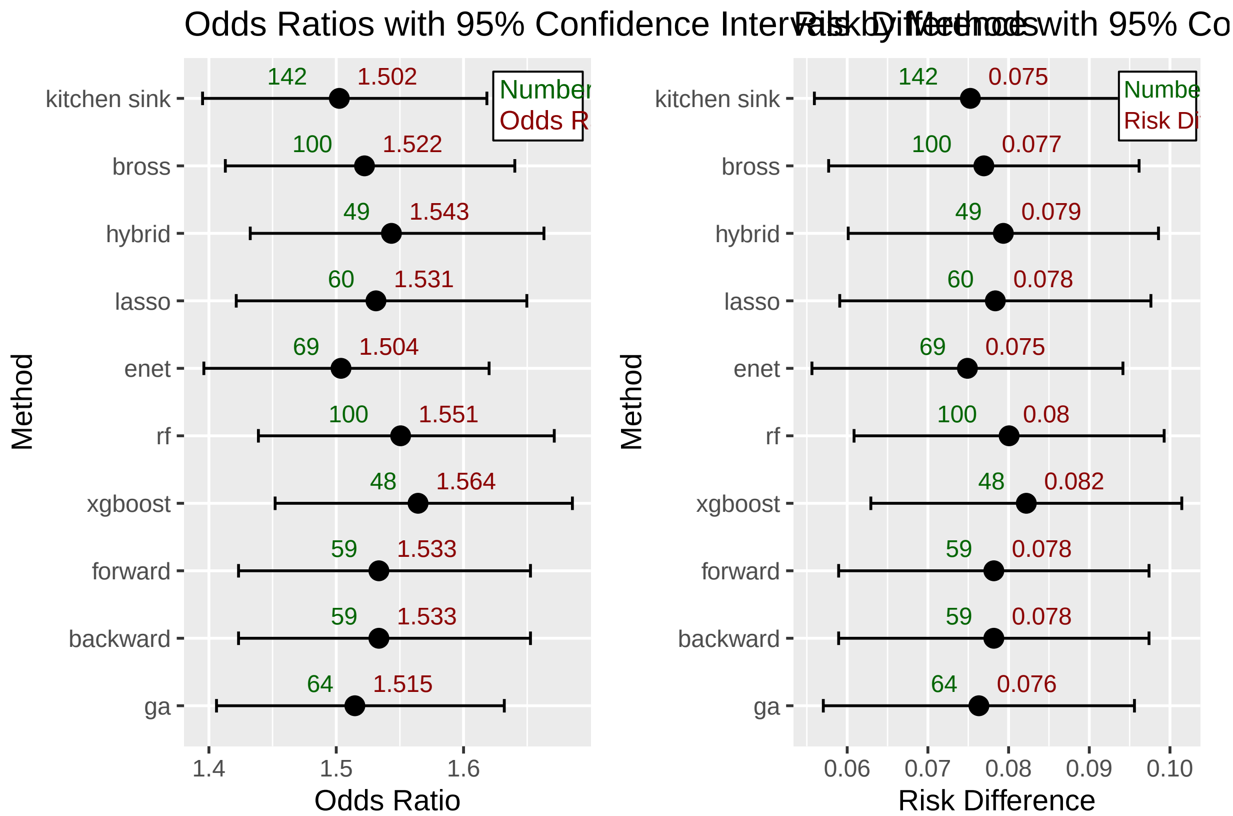

Odds Ratios with 95% Confidence Intervals by Methods
Number of Proxies
Odds Ratios
kitchen sink
bross
hybrid
lasso
enet
rf
xgboost
forward
backward
ga
142
1.502
100
1.522
49
1.543
60
1.531
69
1.504
100
1.551
48
1.564
59
1.533
59
1.533
64
1.515
Method
1.4
1.5
1.6
Odds Ratio




```r
save.image("RealDataAnalysisWorkspace.RData")
```

```r
ggplot2::ggsave("OR_plot.png", plot = OR.plot, device = "png", width = 8, height = 6)
ggplot2::ggsave("RD_plot.png", plot = RD.plot, device = "png", width = 8, height = 6)
```

```
##
## Attaching package: 'patchwork'

## The following object is masked from 'package:MASS':
##
##     area
```




```r
library(ggplot2)
library(patchwork)

# Example: define human-readable names (must match order in 'methods')
method_names <- c(
  "Kitchen Sink", "Bross", "Hybrid", "LASSO", "Elastic Net",
  "Random Forest", "XGBoost", "Forward Selection",
  "Backward Elimination", "Genetic Algorithm"
)

# Create composite labels using real number of proxies
method_labels <- paste0(method_names, " [", num.proxy_all[, 1], "]")

# Rank indices by proxy count (descending order)
rank_idx <- order(num.proxy_all[, 1], decreasing = TRUE)

# Reorder everything based on that rank
method_labels <- method_labels[rank_idx]
#OR_all$method <- factor(method_labels, levels = method_labels)
#RD_all$method <- factor(method_labels, levels = method_labels)

# Apply ordered factor with labels
OR_all$method <- factor(method_labels, levels = rev(method_labels))
RD_all$method <- factor(method_labels, levels = rev(method_labels))

# Plot: OR
```

```r
OR_all
```

```
##                                         method       OR    lower    upper
## sum.ci.OR_ks              Kitchen Sink [142] 1.502410 1.395016 1.618323
## sum.ci.OR_bross                  Bross [100] 1.522207 1.412898 1.640242
## sum.ci.OR_hybrid         Random Forest [100] 1.543358 1.432461 1.663129
## sum.ci.OR_lasso            Elastic Net [69] 1.531204 1.421469 1.649689
## sum.ci.OR_enet       Genetic Algorithm [64] 1.503744 1.396020 1.620046
## sum.ci.OR_rf                     LASSO [60] 1.550569 1.438866 1.671249
## sum.ci.OR_xgboost    Forward Selection [59] 1.564251 1.452001 1.685475
## sum.ci.OR_forward  Backward Elimination [59] 1.533497 1.423263 1.652559
## sum.ci.OR_backward               Hybrid [49] 1.533497 1.423263 1.652559
## sum.ci.OR_ga                    XGBoost [48] 1.514643 1.405970 1.631986
```

```r
OR.plot <- ggplot(OR_all, aes(y = method, x = OR)) +
  geom_point(size = 3, color = "black") +
  geom_errorbarh(aes(xmin = lower, xmax = upper), height = 0.2, color = "black") +
  geom_text(aes(label = round(OR, 2), x = OR + 0.015), vjust = -0.6, size = 3) +
  labs(title = "", x = "Odds Ratio", y = "Method [Chosen number of proxies]") +
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.text.y = element_text(size = 9)
  )



# Plot: RD
RD_all
```

```
##                                         method         RD      lower      upper
## sum.ci.RD_ks              Kitchen Sink [142] 0.07526370 0.05593275 0.09459466
## sum.ci.RD_bross                  Bross [100] 0.07693661 0.05769963 0.09617359
## sum.ci.RD_hybrid         Random Forest [100] 0.07935262 0.06012661 0.09857864
## sum.ci.RD_lasso            Elastic Net [69] 0.07835856 0.05907708 0.09764004
## sum.ci.RD_enet       Genetic Algorithm [64] 0.07490146 0.05563218 0.09417074
## sum.ci.RD_rf                     LASSO [60] 0.08006498 0.06085065 0.09927930
## sum.ci.RD_xgboost    Forward Selection [59] 0.08219667 0.06292909 0.10146424
## sum.ci.RD_forward  Backward Elimination [59] 0.07817236 0.05894149 0.09740323
## sum.ci.RD_backward               Hybrid [49] 0.07817236 0.05894149 0.09740323
## sum.ci.RD_ga                    XGBoost [48] 0.07632127 0.05704192 0.09560062
```

```r
RD.plot <- ggplot(RD_all, aes(y = method, x = RD)) +
  geom_point(size = 3, color = "black") +
  geom_errorbarh(aes(xmin = lower, xmax = upper), height = 0.2, color = "black") +
  geom_text(aes(label = round(RD, 2), x = RD + 0.003), vjust = -0.6, size = 3) +
  labs(title = "", x = "Risk Difference", y = NULL) +
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5),
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank()
```

```
  )


# Combine plots side-by-side
combined_plot2 <- OR.plot + RD.plot +
  plot_layout(ncol = 2, widths = c(1.1, 1))

# Display
combined_plot2
```